Spectral Gradient Field-Guided Bayesian Adaptive Graph Convolution for Hyperspectral Image Classification

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***Abstract*—Graph-based deep learning methods play a significant role in hyperspectral image classification (HSIC). Nonetheless, the paucity of prior information to construct graph model structures continues to severely challenge the classification of images characterized by significant spatial heterogeneity. To achieve effective adaptive construction of graph model structures, this article proposes a spectral-driven self-adapting graph construction framework. A spectral gradient field (SGF) is introduced to quantify the local rate and direction of spectral variation, enabling dynamic adjustment of node connections according to spatial–spectral characteristics. The method adaptively expands or contracts neighborhood scopes based on local spectral gradients, effectively addressing over- and under-segmentation problems in complex regions. Furthermore, a beta distribution is employed to model the probabilistic connectivity between nodes, enhancing the robustness of graph topology. For large hyperspectral scenes, constructing and storing a full graph leads to prohibitive memory consumption and computational cost. To address this issue, we extend the proposed method to a block-wise and pruned Bayesian adaptive graph convolutional network (BP-BAGCN), in which images are divided into nonoverlapping or lightly overlapping spatial blocks; then the local maps are constructed, refined, and inferred independently. Experiments are conducted on three hyperspectral datasets: a benchmark dataset and two newly constructed datasets derived from GF-5 satellite observations over complex coastal wetlands, providing challenging real-world scenarios with rich spectral variability and ecological diversity. Experimental results demonstrate that the proposed method consistently outperforms state-of-the-art approaches under limited labeled samples. Across five independent random seeds, the proposed method achieves average overall accuracies of 98.79% $\pm$ 0.70%, 99.35% $\pm$ 0.13%, and 92.27% $\pm$ 0.65% on the three datasets, respectively, demonstrating both superior classification performance and stable generalization ability.**

***Index Terms*—Adaptive graph structure learning, Bayesian probability modeling, graph convolutional networks (GCNs), hyperspectral image classification (HSIC), spectral gradient field (SGF).**

I. Introduction

THE development of high-resolution spectral imaging sensors, coupled with the mounting demand for detailed classification of land-cover categories, has elevated hyperspectral image classification (HSIC) to a pivotal role within the domain of remote sensing [1], [2]. The manifestation of hyperspectral characteristics is primarily evident in each pixel possessing hundreds of contiguous spectral bands, which provide rich spectral information, enabling fine differentiation between materials with subtle spectral variations [3], [4]. However, due to the high dimensionality of the data and the complex spatial structure of the actual land surface, achieving reliable and scalable classification results poses significant challenges, which become even more severe when labeled samples are limited [5], [6].

Early HSIC methods primarily rely on spectral information and statistical learning models such as random forests [7], [8], support vector machines [9], sparse representations [10], [11], [12], and ensemble learning [13], [14]. While these approaches perform reasonably well in low-complexity scenarios, they are highly sensitive to noise and sample quality due to the failure to fully leverage spatial contextual relationships. To address these limitations, spatial–spectral feature fusion is introduced, incorporating approaches like morphological profiles [15], [16], local binary patterns [17], and superpixel-based representations [18]. Though these methods improve classification accuracy, they often require manual design and tuning for each specific task, lacking adaptability and robustness across diverse scenarios.

With the rapid advancement of deep learning, convolutional neural networks (CNNs) are widely applied to HSIC by integrating spectral and spatial information [19]. 3-D convolutional neural networks (3D-CNNs), in particular, demonstrate

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formidable performance by directly learning hierarchical spatial–spectral representations from data. Nevertheless, CNN-based methods inherently rely on fixed grid structures and local receptive fields. Their feature modeling capabilities are primarily constrained by kernel size and network depth, which limit their ability to effectively model long-range spatial dependencies and irregular spatial interactions. The Transformer architecture [20], based on self-attention mechanisms, is introduced into hyperspectral image analysis to overcome the limitations of convolutional operations in global modeling. Meanwhile, recently emerging sequence modeling methods based on state-space models (SSMs) also garner increasing attention, such as the Mamba architecture. Unlike Transformers, Mamba employs a linear-time-complexity selective state update mechanism that significantly reduces computational and memory overhead while preserving long-range dependency modeling capabilities, offering a novel approach for modeling high-dimensional spectral sequences [21]. Graph neural networks (GNNs) offer flexible solutions for modeling non-Euclidean structured data, garnering significant attention in hyperspectral remote sensing image analysis [22]. Graph-based models enable adaptive message passing across irregular neighborhoods by representing pixels as nodes and encoding spatial or spectral relationships as edges. Among these, graph convolutional networks (GCNs) demonstrate substantial potential in HSIC. Unlike CNN methods that rely on explicit spatial feature extraction, GCNs construct graph structures based on simple spatial neighborhood relationships defined by local connectivity. Spatial information is not encoded through artificially designed features but is implicitly modeled through graph structure message passing.

Most existing GCN-based methods still rely on fixed or heuristically defined adjacency matrices. However, such static graph structures often fail to capture complex class boundaries and may propagate misleading information across heterogeneous regions. To address this issue, adaptive graph modeling strategies are required to dynamically adjust the graph structure based on data-driven evidence and initial model predictions.

This article proposes a Bayesian adaptive graph convolutional network (BAGCN) for HSIC based on a spectral gradient field (SGF). This network explicitly integrates gradient field information and Bayesian probability into a graph convolutional learning framework. From a physical perspective, spectral variations across continuous bands contain crucial information about material composition and transitions. The SGF provides a distinct representation of spectral variation patterns. Unlike raw spectral values, the SGF emphasizes local spectral changes, captures whether spectral values exhibit ascending or descending trends, and quantifies the intensity of spectral changes. Bayesian probability modeling refines graph structures based on the observed evidence. By treating edge existence or strength as random variables and incorporating posterior uncertainty into adjacency estimation, it suppresses unreliable connections while enhancing informational content.

The proposed method does not rely on fixed adjacency matrices. Instead, it models each graph edge using a Bayesian posterior distribution, updating its parameters by combining spectral consistency from SGF with soft class probability estimates generated by GCN. In this manner, spectral gradient information not only serves as input features but also actively guides the adjustment of adaptive weights for message passing between neighboring pixels. To enhance scalability, a block-wise and pruned Bayesian adaptive graph convolutional network (BP-BAGCN) is further proposed. Hyperspectral images are partitioned into nonoverlapping or lightly overlapping spatial blocks. Within these blocks, local graphs are independently constructed, refined, and inferred, significantly reducing computational complexity without compromising classification performance. Predictions from overlapping blocks are subsequently merged to yield a globally consistent classification map. The main contributions of this work can be summarized as follows.

1) A novel Bayesian adaptive graph convolutional framework is proposed, which dynamically optimizes graph adjacency by jointly modeling SGF consistency and class posterior evidence, enabling data-driven and uncertainty-aware message propagation.
2) SGF information is systematically incorporated into graph construction and adjacency refinement, providing physically interpretable guidance for message passing beyond conventional spectral similarity measures.
3) To address large-scale hyperspectral scenes, a block-wise graph inference strategy combined with Bayesian posterior-based edge pruning is developed, significantly reducing memory footprint and computational complexity while preserving classification accuracy and spatial consistency.
4) The proposed method is introduced and validated on one benchmark dataset and two newly constructed GF-5 hyperspectral datasets covering coastal wetland areas in Yancheng City, Jiangsu Province, enabling a more comprehensive evaluation of model generalization in practical scenarios.

The remainder of this article is organized as follows. A comprehensive review of existing Transformer-, Mamba-, and graph-based methods is provided in Section II. Section III reviews related theoretical work on graph-based message passing, spectral gradient representations, and Bayesian probabilistic modeling. Section IV details the proposed Bayesian adaptive graph convolutional framework and its block-wise extension. Experimental results and analysis are reported in Section V, while conclusions are presented in Section VI.

II. Related Work

This section reviews recent HSIC methods, including Transformer-based architectures, Mamba-based approaches, and graph-based models.

A. Transformer-Based Architectures for HSIC

By explicitly modeling global dependencies between arbitrary pixel pairs, Transformers exhibit strong expressive power in capturing long-range spatial–spectral correlations. Hong et al. [23] proposed SpectralFormer, which is among the earliest attempts to rethink HSI classification from a spectral sequence modeling perspective, where groupwise spectral embeddings and cross-layer skip connections are designed to

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preserve spectrally local dependencies and mitigate information loss during deep propagation. To enhance spectral–spatial interaction, Peng et al. [24] introduced a dual-branch Transformer with spatial–spectral cross-attention, enabling mutual feature refinement while alleviating computational redundancy through weighted sharing mechanisms.

Several works further integrated convolutional inductive biases into Transformer frameworks. Yang et al. [25] embedded spectral-adaptive 3-D convolutions into the Transformer projection stage to capture subtle spectral discrepancies and local spatial context, while Song et al. [26] proposed a parallel design combining multihead self-attention and local spatial perception, together with a bidirectional interaction mechanism to reinforce complementary spatial–spectral cues. Similarly, Yu et al. [27] employed a multilevel Transformer encoder–decoder architecture with positional embedding to jointly learn hierarchical spectral–spatial representations. To address multiscale feature modeling, Chen et al. [28] proposed a scale-interaction Transformer (SiT) encoder that explicitly introduces scale sets into the attention mechanism, enabling global dependency modeling from multiscale spectral and spatial perspectives. MSSCFormer [29] further investigated multigranularity representation learning by integrating intraclass spectral attention and global–local convolutional Transformers to reduce intraclass spectral variability. In addition, Bai et al. [30] introduced a multibranch Transformer architecture with spatial sequence attention and spectral feature extraction, aiming to suppress irrelevant background interference and improve boundary pixel classification.

Considering the large parameter size and overfitting risk of vanilla Transformers, several lightweight and efficient variants have been proposed. Zhao et al. [31] adopted groupwise separable convolution and attention to reduce computational cost while preserving local–global feature representation. Zhang et al. [32] designed lightweight self-attention modules combined with convolutional tokenization to improve efficiency and generalization under limited training samples. Mei et al. [33] introduced grouped pixel embedding and hierarchical Transformer structures to confine attention dispersion and enhance local spectral–spatial consistency.

Recently, advanced fusion and structural modeling strategies have been explored. Xu et al. [34] bridged the CNN and Transformer through cross-attention fusion to jointly exploit local inductive bias and global dependency modeling. Roy et al. [35] incorporated learnable spectral–spatial morphological operations into the Transformer attention mechanism to strengthen structural representation. Zhang et al. [36] avoided explicit sequence flattening by directly operating on HSI tensors via tensor self-attention, preserving inherent spatial–spectral structural relationships. Additionally, Jia et al. [37] introduced a self-supervised pretraining strategy with center-masked modeling to improve representation learning efficiency and robustness.

B. Mamba-Based Approaches for HSIC

Early attempts explored the feasibility of adapting 1-D Mamba models to hyperspectral data. Wang et al. [38] investigated noncausal spatial–spectral modeling by introducing multidirectional local spatial scanning and bidirectional spectral scanning mechanisms, enhancing local feature perception while preserving linear complexity. Similarly, Zhou et al. [39] proposed a centralized cross-scan strategy and tokenized Mamba encoder to improve patch-wise semantic aggregation and robustness under limited training samples.

To overcome the limitations of flattening hyperspectral cubes into 1-D sequences, several works explicitly redesigned token generation and scanning strategies. He et al. [40] introduced a spectral–spatial token generation module and a 3-D selective scanning mechanism, enabling pixel-wise scanning along spectral and spatial dimensions to extract global spectral–spatial semantics. They [41] further proposed an interval group SSM mechanism that partitions high-dimensional features into nonoverlapping groups and performs directional scanning within each group, achieving multidirectional global modeling with reduced redundancy and computational cost.

Several studies focused on enhancing structural and multiscale representation learning. Ahmad et al. [42] incorporated morphological operations such as dilation and erosion into the Mamba framework to capture structural and shape-aware spatial–spectral features. Zhang et al. [43] further explored time–frequency representations by integrating spectral and spatial wavelet decomposition into Mamba blocks for enhanced spectral–spatial fusion. To improve local–global feature complementarity, hybrid architectures combining Mamba with convolution or Transformer components have been proposed. Sheng et al. [44] adopted a parallel Mamba–CNN design to jointly extract global spectral–spatial dependencies and local contextual details, while He et al. [45] integrated Mamba experts for long-range modeling with Transformer experts for short-range dependency capture across multiple scales. Liang et al. [46] introduced a Mamba-like linear attention mechanism inspired by Transformers to improve global modeling capability while maintaining linear complexity.

Lately, efforts have been made to incorporate graph modeling and adaptive structure learning into Mamba-based frameworks. Yang et al. [47] combined efficient Mamba encoding with spatial graph convolution to adaptively aggregate multihop spatial context, highlighting the potential of integrating SSMs with explicit relational reasoning for hyperspectral image analysis.

C. Graph-Based Models for HSIC

Early studies demonstrated that representing hyperspectral images as graphs enables more effective exploitation of spatial context under limited training samples. Mou et al. [48] are among the first to formulate HSI classification in a semi-supervised graph learning paradigm, where the entire image is treated as a nonlocal graph to jointly leverage labeled and unlabeled samples. Similarly, Wan et al. [49] refined graph connectivity and edge weights during layer-wise propagation, allowing long-range contextual relations to be gradually captured through successive graph convolutions.

To alleviate the reliance on fixed graph structures, several works explored adaptive graph learning mechanisms.

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Yang et al. [50] proposed a dynamic graph integration strategy across different network layers, where learned graphs were adaptively fused with the initial adjacency to enhance representation robustness while mitigating over-smoothing through residual and identity mappings. Chen et al. [51] further unified graph construction and feature learning in an end-to-end framework, embedding a Siamese network to automatically learn and update graph structures in a task-driven manner.

Beyond pixel-level graphs, superpixel-based and multiscale graph representations have been widely adopted to balance computational efficiency and spatial semantic consistency. Liu et al. [52] extended this idea to multilevel superpixel graphs and proposed graph U-Net-like pooling and unpooling operations, enabling hierarchical feature learning across multiple spatial scales. To further enhance structural expressiveness, several studies incorporated more advanced graph formulations. Jia et al. [53] proposed a graph-in-graph representation to jointly encode intrasuperpixel and intersuperpixel relationships, enabling hierarchical aggregation of local and global object characteristics. Zheng et al. [54] introduced Kolmogorov–Arnold network theory into graph convolution to improve model interpretability and scalability by replacing conventional MLP-based message passing with spline-based functional mappings.

More recently, hypergraph-based models have been developed to explicitly capture high-order relationships among multiple pixels. Wang et al. [55] constructed diversified hyperedges guided by superpixel segmentation and fused multiple hypergraph structures to enrich global topology representation under limited samples. Yang et al. [56] further extended graph convolution into the tensor domain by proposing a tensorized high-order hypergraph representation, enabling efficient extraction of high-order discriminative information via tensor decomposition, while adopting minibatch training to address memory bottlenecks in large-scale scenes.

III. Theoretical Background

A. Graph Message Passing

GNNs are predicated on the paradigm of message passing, wherein node representations are iteratively updated through information aggregation from neighboring nodes. This framework offers a unifying perspective on various graph-based learning models and has been extensively adopted in non-Euclidean data analysis.

The majority of GNNs can be formulated under the message passing neural network (MPNN) framework. Given a graph, the hidden representation of node i at layer $l+1$ is updated as

$$h_i^{(l+1)} = \phi^{(l)}\left(h_i^{(l)}, \bigoplus_{j\in N(i)} \psi^{(l)}\left(h_i^{(l)}, h_j^{(l)}, e_{ij}\right)\right) \tag{1}$$

where $\phi(\cdot)$ is the update function, node j is the index of an adjacent node of node i, $N(i)$ is the neighborhood set, $\psi(\cdot)$ denotes the message function, $\oplus$ represents a permutation-invariant aggregation operator (e.g., sum or mean), and e_{ij} is the edge feature between specified nodes. This formulation highlights that graph structure directly governs how information is propagated, making the quality of message passing highly dependent on the reliability of graph edges.

Early GCNs are derived from spectral graph theory, where convolution is defined in the Fourier domain of graphs. The widely used GCN model simplifies spectral convolution as

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{2}$$

where $\mathbf{H}^{(l+1)}$ is the node features matrix from layer $l+1$, A is the adjacency matrix of the graph, and $\tilde{\mathrm{A}} = \mathrm{A} + \mathrm{I}$ is the augmented adjacency matrix. $\tilde{\mathbf{D}}$ denotes the corresponding degree matrix with diagonal entries $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$. Besides, the symmetric normalization $\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$ is adopted to stabilize the feature propagation process. $\mathbf{W}^{(l)}$ represents the trainable weight matrix, and $\sigma(\cdot)$ denotes a nonlinear activation function such as ReLU. Despite its efficiency, this formulation assumes a fixed and deterministic adjacency matrix, implicitly treating all neighboring nodes as equally reliable information sources.

To mitigate the rigidity of fixed graph structures, attention mechanisms have been introduced to adaptively weight messages. Graph attention networks (GATs) compute attention coefficients as

$$\alpha_{ij} = \frac{\exp\left(a\left(h_i, h_j\right)\right)}{\sum_{k\in N(i)} \exp\left(a\left(h_i, h_k\right)\right)} \tag{3}$$

which measures the importance of node j to node i. h_k denotes the feature vectors while $N(i)$ represents the set of neighbors of node i. The function $a(\cdot)$ is a learnable attention mechanism used to compute the pairwise importance between nodes. The exponential function and normalization over $k \in N(i)$ implement a Softmax operation, ensuring that the attention coefficients of all neighbors sum to one. Through this mechanism, the GAT adaptively assigns different weights to neighboring nodes, allowing the model to capture the varying importance of neighborhood information during feature aggregation.

B. SGF Representation

The SGF provides a physically interpretable representation of spectral variation by characterizing local changes along the spectral dimension. Given the spectral reflectance vector of pixel i

$$x_i = \left[x_i(1), x_i(2), \ldots, x_i(B)\right]^{\mathrm{T}}. \tag{4}$$

where B is the total number of pixel bands, and the SGF is defined as the first-order derivative with respect to the spectral band index b

$$g_i(b) = \frac{\partial x_i(b)}{\partial b}. \tag{5}$$

In practice, the SGF is approximated using finite differences

$$g_i(b) \approx \frac{x_i(b+1) - x_i(b-1)}{2} \tag{6}$$

with forward and backward differences applied at boundary bands.

The sign of the spectral gradient reflects the direction of spectral variation; a positive gradient $g_i(b) > 0$ indicates a rising spectral trend, corresponding to increasing reflectance intensity across adjacent spectral bands. Negative gradient $g_i(b) < 0$ indicates a descending spectral trend, corresponding to decreasing reflectance intensity.

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From a physical perspective, these gradient directions are closely related to material-dependent spectral behaviors, such as absorption features, reflectance transitions, and spectral slope changes. Materials with similar absolute reflectance values may exhibit distinct gradient patterns, making SGF particularly effective for discriminating spectrally similar classes.

Beyond directional information, the magnitude of the spectral gradient quantifies the intensity of spectral variation, which is defined as

$$\|g_i\|_2 = \left(\sum_{b=1}^{B} g_i(b)^2 \right)^{\frac{1}{2}}. \tag{7}$$

A larger gradient magnitude indicates a greater spectral change across bands, often corresponding to sharp absorption edges or rapid spectral transitions. Conversely, a smaller magnitude reflects spectrally smooth regions with limited variation. The gradient magnitude thus serves as a robust measure of spectral activity, which is less sensitive to absolute reflectance levels and more indicative of intrinsic material properties.

Compared with conventional spectral descriptors, spectral derivatives mainly characterize local band-wise variations, band-difference features emphasize pairwise spectral contrast, and morphological profiles focus on multiscale spatial structures. SGF represents spectral responses as a continuous gradient field that jointly encodes the direction and magnitude of spectral variation across bands. Consequently, SGF provides complementary spectral and structural information, leading to more discriminative hyperspectral representations.

C. Probabilistic and Bayesian Model Learning

Probabilistic and Bayesian learning frameworks provide a principled methodology for modeling uncertainty in machine learning systems. In contrast to deterministic models, which produce point estimates, Bayesian approaches treat model components as random variables and infer posterior distributions conditioned on observed data.

Given observed data D, Bayesian learning aims to infer the posterior distribution of model parameters θ through Bayes' rule

$$p(\theta|D) = \frac{p(D|\theta)\, p(\theta)}{p(D)} \tag{8}$$

where $p(\theta)$ denotes the prior distribution encoding prior knowledge, and $p(D|\theta)$ is the likelihood function.

In probabilistic graph modeling, the presence or strength of an edge e_{ij} is treated as a random variable

$$e_{ij} \sim p\left(e_{ij}\right) \tag{9}$$

where the prior distribution reflects prior beliefs about node connectivity.

A common choice for modeling edge existence or confidence is the Beta distribution

$$p\left(e_{ij}\right) \sim \mathrm{Beta}(\alpha, \beta) \tag{10}$$

are shape parameters. Given observed evidence ε_{ij}, the which provides a flexible prior over probabilities, α and β posterior distribution of e_{ij} can be updated according to Bayes' rule as

$$p\left(e_{ij}|\varepsilon_{ij}\right) \propto p\left(\varepsilon_{ij}|e_{ij}\right) p\left(e_{ij}\right) \tag{11}$$

Algorithm 1 Bayesian Adaptive Graph Convolutional Network (BAGCN)

Input:

(1) Hyperspectral image $\mathbf{X} \in \mathrm{R}^{H \times W \times B}$
(2) Ground-truth labels $\mathbf{Y}$
(3) Bayesian prior parameters (α, β)
(4) Spectral gradient field operator $\mathcal{G}(\cdot)$

Procedure:

1. Compute spectral gradient field: $\mathbf{G} = \mathcal{G}(\mathbf{X})$
2. Construct node features: $\mathbf{F} = [\mathbf{X}, \mathbf{G}]$
3. Initial Graph Construction: Construct initial spatial adjacency matrix $\mathbf{A}$ using grid-based neighborhood
4. Bayesian Adaptive Graph Refinement:
   for each edge e_{ij}:
   Compute the probability of same class p_{ij} via Eq. (21)
   Compute SGF-based similarity s_{ij} via Eq. (22)
   Compute class-consistency evidence ε_{ij} via Eq. (23)
   Update posterior edge confidence via Eq. (24)
   End for
   Obtain adaptive adjacency matrix $\hat{\mathbf{A}}$ via Eq. (25)
5. Graph Convolutional Inference:
   $\tilde{\mathbf{A}} = \hat{\mathbf{A}} + \mathbf{I}$
   $\bar{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}}$
   $\mathbf{H}^{(1)} = \sigma(\bar{\mathbf{A}}\mathbf{X}\mathbf{W}^{(0)})$
   $\mathbf{H}^{(2)} = \bar{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}^{(1)}$
6. Prediction:
   Predict class labels $\hat{\mathbf{Y}}$ via Softmax

Output: predicted label map $\hat{\mathbf{Y}}_{\mathrm{BAGCN}}$

where $p(\varepsilon_{ij}|e_{ij})$ represents the likelihood function, and the symbol $\propto$ denotes proportionality. This Bayesian formulation allows graph connectivity to be refined iteratively based on data-driven evidence, enabling adaptive graph learning under uncertainty.

IV. Proposed Method

In this section, we present a Bayesian adaptive graph convolutional network guided by SGF for HSIC, followed by an efficient block-wise and pruned inference extension designed for large hyperspectral scenes. As illustrated in Fig. 1, the overall framework of BAGCN integrates spectral feature extraction, SGF modeling, and graph construction into a unified pipeline for inference. Fig. 2 details the internal architecture of BAGCN, including feature extraction and construction, initial graph construction, and Bayesian graph refinement. The detailed structure of the BP-BAGCN extension, including block-wise graph construction and Bayesian graph edge pruning, is illustrated in Fig. 3.

Denote the HSI cube as

$$X \in \mathrm{R}^{H \times W \times B} \tag{12}$$

where H and W represent the spatial dimensions, and B is the number of spectral bands. Let $x_i \in \mathrm{R}^B$ denote the spectral vector of the ith pixel, and $y_c \in \{1, 2, \ldots, C\}$ be its reference class label when available.

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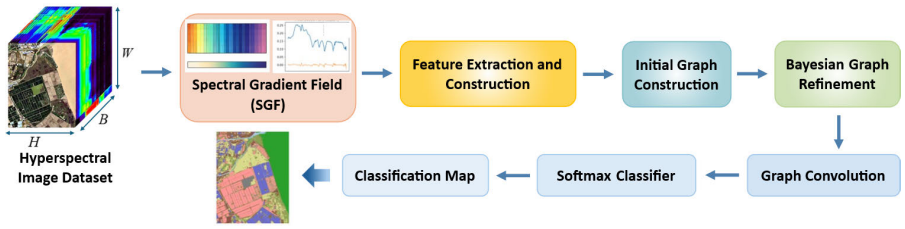


Fig. 1. Overview of the proposed SGF-guided BAGCN framework for HSIC. The framework extracts spectral and spectral-gradient representations from the input hyperspectral cube, followed by graph construction and Bayesian adaptive edge refinement. The refined graph is processed through graph convolution to generate node representations and final classification maps.

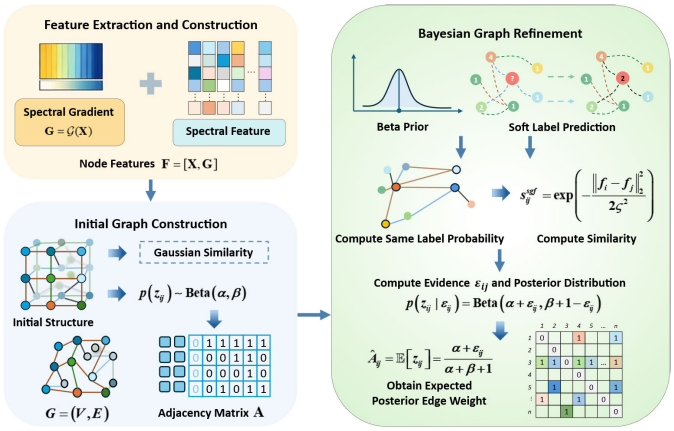


Fig. 2. Detailed architecture of the BAGCN module. SGF features are computed to characterize local spectral variation and fused with spectral representations. Pairwise feature similarity and prediction consistency are incorporated into Bayesian posterior estimation to adaptively update graph connectivity, followed by graph convolution for neighborhood information aggregation.

A. SGF-Guided Bayesian Adaptive Graph Convolution

Each pixel x_i in X corresponds to a node in a graph, resulting in $N = H \times W$ nodes, $i = 1, 2, \ldots, N$. For each pixel spectrum $x_i = [x_i^{(1)}, \ldots, x_i^{(B)}]$, the SGF is defined as a discrete spectral derivative along the wavelength dimension

$$g_i^{(b)} = \begin{cases} x_i^{(b+1)} - x_i^{(b)}, & b = 1 \\ \dfrac{x_i^{(b+1)} - x_i^{(b-1)}}{2}, & 1 < b < B \\ x_i^{(b)} - x_i^{(b-1)}, & b = B. \end{cases} \tag{13}$$

This formulation captures local spectral slope changes, which are closely related to material absorption characteristics and spectral shape variations. SGF is derived purely from spectral variations along the wavelength dimension, without involving any spatial convolution or neighborhood aggregation.

Additionally, the SGF magnitude is computed as

$$m_i = \|g_i\|_2 \tag{14}$$

providing a compact scalar measure of the overall spectral variation strength at pixel x_i.

Then, PCA is applied separately to spectral vectors and SGFs to reduce redundancy; the final node feature vector is constructed as

$$f_i = \left[\mathrm{PCA}\left(x_i\right), \mathrm{PCA}\left(g_i, \hat{m}_i\right)\right] \quad i = 1, 2, \ldots, N. \tag{15}$$

$\hat{m}_i$ denotes the normalized SGF magnitude. SGF is not merely regarded as an ancillary feature; instead, it enriches node representations and guides adaptive graph structure refinement. The proposed method employs an 8-neighborhood grid to construct the graph, where each node corresponds to a pixel and edges encode local spatial adjacency. The initial adjacency matrix $\mathbf{A}$ is defined as

$$A_{ij} = \begin{cases} 1, & \text{if pixels } i \text{ and } j \text{ are spatial neighbors} \\ 0, & \text{othrtwise} \end{cases}$$
$$i, j = 1, 2, \ldots, Ni \neq j. \tag{16}$$

Self-loops are added, and symmetric normalization is applied

$$\tilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}} \left(\mathbf{A} + \mathbf{I}\right) \mathbf{D}^{-\frac{1}{2}} \tag{17}$$

where $\mathbf{D}$ is the degree matrix. This initial graph encodes pure spatial proximity, without assuming spectral homogeneity. For each candidate edge e_{ij}, define a latent variable

$$z_{ij} \in \{0, 1\} \quad i, j = 1, 2, \ldots, Ni \neq j \tag{18}$$

indicating whether nodes i and j should exchange information.

A Beta prior is imposed

$$p\left(z_{ij}\right) \sim \mathrm{Beta}\left(\alpha, \beta\right). \tag{19}$$

After an initial GCN training stage, soft label predictions are obtained

$$\mathbf{p}_i = \mathrm{Softmax}\left(\mathbf{h}_i\right) \quad i = 1, 2, \ldots, N \tag{20}$$

where $\mathbf{h}_i$ denotes the logit output of the GCN.

The probability that two nodes belong to the same class is estimated as

$$p_{ij}^{cls} = \mathbf{p}_i^{\mathrm{T}} \mathbf{p}_j \quad i, j = 1, 2, \ldots, N \quad i \neq j. \tag{21}$$

Meanwhile, SGF consistency is quantified using a Gaussian kernel

$$s_{ij}^{sgf} = \exp\left(-\frac{\left\|f_i - f_j\right\|_2^2}{2\varsigma^2}\right)$$
$$i, j = 1, 2, \ldots, N \quad i \neq j \tag{22}$$

where ς controls the sensitivity to spectral variation differences. The Gaussian kernel provides a smooth and continuous measure of similarity, ensuring that closer nodes in the feature space have stronger connections. This property is particularly suitable for hyperspectral data, where subtle spectral differences are critical for classification. Moreover, the exponential

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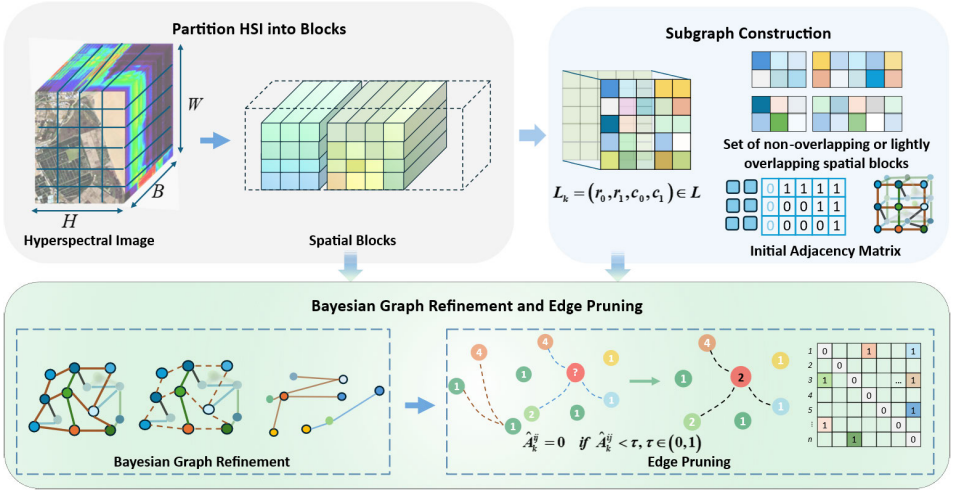


Fig. 3. Detailed illustration of the BP-BAGCN. Large hyperspectral scenes are divided into local spatial blocks for scalable graph construction and inference. Posterior edge probabilities are further used to remove weak connections, reducing memory consumption and computational complexity while preserving discriminative graph structures.

decay form enhances locality while suppressing noisy or weak correlations, leading to more stable graph construction.

The combined evidence is defined as

$$\varepsilon_{ij} = p_{ij}^{cls} \cdot s_{ij}^{sgf} \quad i, j = 1, 2, \ldots, N \quad i \neq j. \tag{23}$$

The multiplicative fusion acts as a consensus-based confidence mechanism that reinforces graph connectivity only when both information sources indicate high reliability. Such a design suppresses noisy edge propagation and improves structural stability during adaptive graph refinement and does not imply conditional independence between semantic agreement and SGF similarity. Alternative formulations (e.g., additive, learned-weighting, and uncertainty-adaptive fusion) either weaken the conservative edge-selection principle or increase model complexity.

The posterior distribution becomes

$$p\left(z_{ij}|\varepsilon_{ij}\right) = \text{Beta}\left(\alpha + \varepsilon_{ij}, \beta + 1 - \varepsilon_{ij}\right). \tag{24}$$

The expected posterior edge weight is

$$\hat{A}_{ij} = \text{E}\left[z_{ij}\right] = \frac{\alpha + \varepsilon_{ij}}{\alpha + \beta + 1}. \tag{25}$$

After normalization, the refined adjacency matrix $\tilde{\mathbf{A}} = \hat{\mathbf{A}} + \mathbf{I}$ is used for subsequent GCN inference. Given the refined adjacency matrix $\tilde{\mathbf{A}}$, a two-layer GCN is employed

$$\bar{\mathbf{A}} = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-\frac{1}{2}} \tag{26}$$

$$\mathbf{H}^{(1)} = \sigma\left(\bar{\mathbf{A}}\mathbf{X}\mathbf{W}^{(0)}\right) \tag{27}$$

$$\mathbf{H}^{(2)} = \bar{\mathbf{A}}\mathbf{H}^{(1)}\mathbf{W}^{(1)} \tag{28}$$

where $\mathbf{W}^{(0)}, \mathbf{W}^{(1)}$ are trainable weights, and $\sigma(\cdot)$ denotes ReLU activation. The final classification results are obtained by applying a Softmax function to the output logits of the network.

Unlike classical Beta-Bernoulli conjugate inference that relies on discrete edge observations, the proposed framework introduces a continuous evidence variable derived from spectral gradient consistency and soft class agreement. This evidence is interpreted as a soft confidence signal describing the likelihood that two nodes should remain connected. Accordingly, posterior updating follows a generalized evidence accumulation mechanism rather than exact Bernoulli observation counting. The Bayesian component serves primarily as an uncertainty-aware graph adaptation mechanism rather than uncertainty quantification for final class prediction. The Bayesian formulation enables graph connectivity to be adaptively estimated according to data-driven evidence under uncertainty. In the implementation adopted, the posterior edge probabilities are computed once to generate the refined graph structure, while subsequent optimization is performed through standard network parameter learning rather than repeated outer-loop graph reconstruction.

Unlike conventional pseudo-label or self-training strategies that directly update graph structures using deterministic class assignments, the proposed refinement mechanism adopts uncertainty-aware edge updating based on continuous evidence. Class information is introduced through soft posterior agreement rather than hard labels, allowing uncertain predictions to contribute weak evidence instead of enforcing incorrect graph evolution. SGF similarity acts as an independent structural constraint that regularizes edge adaptation. Consequently, graph connectivity is strengthened only when both class confidence and spectral-gradient consistency support the update, which alleviates potential error reinforcement caused by noisy initial predictions.

B. Block-Wise Bayesian Graph Convolution and Graph Pruning

For large hyperspectral scenes, constructing and storing a full graph leads to prohibitive memory consumption and computational cost. To address this issue, the proposed method is further extended into a block-wise inference framework.

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The HSI is divided into a set of nonoverlapping or lightly overlapping spatial blocks

$$L_k = (r_0, r_1, c_0, c_1) \in L \tag{29}$$

r and c represent the number of rows and columns, respectively. Let $\mathbf{X}_k \in \mathbb{R}^{N_k \times d}$ denote the block-wise feature matrix, where $N_k = L_h \times L_w$ is the number of pixels in the kth block, and d denotes the feature dimensionality. The feature vector of each pixel is composed of the original spectral bands and the corresponding SGF responses.

For each block, SGF representations and spectral features are computed locally. A local spatial graph is constructed based on pixel neighborhood connectivity, and Bayesian adjacency refinement is subsequently performed to adaptively update edge weights by exploiting block-level spectral-gradient consistency and soft class probability estimates. Specifically, each edge weight e_k^{ij} is modeled as a random variable following a Beta distribution:

$$p\left(e_k^{ij}\right) \sim \text{Beta}\,(\alpha, \beta) \tag{30}$$

where the parameters α and β encode prior beliefs about the reliability of message propagation between nodes i and j.

Calculate the evidence value and posterior distribution weights for each block using the same procedure as BAGCN. Then, edges with posterior weights below a threshold τ are pruned

$$\hat{A}_k^{ij} = 0 \quad if\ \hat{A}_k^{ij} < \tau,\ \tau \in (0, 1)\,. \tag{31}$$

Following the refinement of the local graph structure, GCN inference is conducted independently within each block, enabling localized message passing while significantly reducing computational and memory overhead:

$$\mathbf{H}_k^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}_k^{-\frac{1}{2}} \tilde{\mathbf{A}}_k \tilde{\mathbf{D}}_k^{-\frac{1}{2}} \mathbf{H}_k^{(l)} \mathbf{W}_k^{(l)}\right) \tag{32}$$

where $\tilde{\mathbf{A}}_k = \hat{\mathbf{A}}_k + \mathbf{I}$ and $\mathbf{W}_k^{(l)}$ are shared across all blocks. For overlapping regions in block-wise inference, majority voting is employed to aggregate multiple predictions, yielding a globally consistent classification map while preserving local adaptivity. Other fusion strategies such as probability averaging or weighted voting can also be adopted, but majority voting offers a favorable tradeoff between simplicity and stability.

The proposed BP-BAGCN adopts block-wise processing as an efficient approximation for scalable graph learning in large hyperspectral scenes. By preserving global spectral-gradient representations before partitioning, refining edge confidence through Bayesian adaptation, and pruning low-confidence connections, the framework reduces computational cost while maintaining robust graph structure modeling and classification performance.

C. Memory Consumption and Computational Complexity Analysis

To evaluate the scalability advantage of the proposed framework, this section analyzes both computational complexity and memory consumption of BAGCN and its block-wise extension BP-BAGCN.

Unlike dense graph modeling, the proposed method adopts sparse local connectivity. Let $|E|$ represent the total number of graph edges. Since the initial graph uses an eight-neighborhood topology, $|E| \approx 8N$.

Algorithm 2 Block-Wise and Pruned Bayesian Adaptive Graph Convolutional Network

Input:

(1) Hyperspectral image $\mathbf{X} \in \mathrm{R}^{H \times W \times B}$
(2) Ground-truth labels $\mathbf{Y}$
(3) Bayesian prior parameters (α, β)
(4) Spectral gradient field operator $\mathcal{G}(\cdot)$
(5) Set parameters: block size B_s, overlap margin δ, pruning threshold τ

Main Loop:

1. Partition HSI into k possibly overlapping blocks:
$L_k = (r_0, r_1, c_0, c_1) \in L$
for each block:
1.1 Subgraph construction
Extract block data $\mathbf{X}_k$ and labels $\mathbf{Y}_k$
Compute block-wise SGF: $\mathbf{G}_k = \mathcal{G}(\mathbf{X}_k)$
Construct node features: $\mathbf{F}_k = [\mathbf{X}_k, \mathbf{G}_k]$
Build initial adjacency matrix $\mathbf{A}_k$
1.2 Bayesian Graph Refinement:
for each edge e_k^{ij}
Estimate posterior edge confidence via Eq. (30)
End for
1.3 Posterior-based Edge Pruning according to Eq. (31)
1.4 Local GCN Inference:
Apply GCN on pruned subgraph:
$\tilde{\mathbf{A}}_k = \hat{\mathbf{A}}_k + \mathbf{I}$
$\bar{\mathbf{A}}_k = \tilde{\mathbf{D}}^{-\frac{1}{2}} \tilde{\mathbf{A}}_k \tilde{\mathbf{D}}^{-\frac{1}{2}}$
$\mathbf{H}_k^{(1)} = \sigma(\bar{\mathbf{A}}_k \mathbf{X}_k \mathbf{W}_k^{(0)})$
$\mathbf{H}_k^{(2)} = \bar{\mathbf{A}}_k \mathbf{H}_k^{(1)} \mathbf{W}_k^{(1)}$
Obtain block-level predictions $\widehat{\mathbf{Y}}_k$ via Softmax
End for
2. Block Fusion:
Aggregate predictions from overlapping blocks via voting averaging.

Output: final classification map $\hat{\mathbf{Y}}_{\text{BP-BAGCN}}$

1) Complexity of BAGCN: The memory consumption mainly consists of three parts: node feature storage, graph adjacency storage, and intermediate graph convolution activations. Feature storage requires $M_F = O(Nd)$, adjacency storage under sparse representation requires $M_A = O(|E|)$. For graph convolution, intermediate propagation features require $M_g = O(Ndh_l)$. Therefore, total memory complexity becomes $M_{BAGCN} = O(Nd + |E| + Ndh_l) \approx O(N(8 + d + dh_l))$.

For computational complexity, one graph convolution layer can be expressed as $H^{(l+1)} = \sigma(\bar{A}H^{(l)}W^{(l)})$, where $\bar{A} \in \mathbb{R}^{N \times N}$, $H^{(l)} \in \mathbb{R}^{N \times d}$, and $W^{(l)} \in \mathbb{R}^{d \times h_l}$. Since sparse multiplication only traverses existing edges, edge aggregation complexity becomes $T_A = O(|E|h_l)$ and feature transformation requires $T_W = O(Ndh_l)$. $T_{BAGCN} = O(|E|h_l + Ndh_l) \approx O(Nh_l(8 + d))$, which grows approximately linearly with image size.

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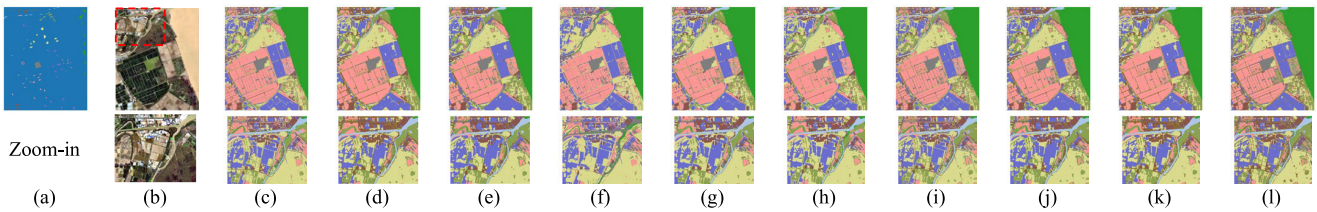


Fig. 4. Classification maps and zoomed-in boundary region on the Yaowa Port dataset. (a) Ground truth. (b) False-color image. (c) miniGCN [57]. (d) GiGCN [53]. (e) GKACN [54]. (f) ViT [20]. (g) SF [23]. (h) SSFTT [58]. (i) MiM [39]. (j) SSMamba [44]. (k) BAGCN. (l) BP-BAGCN.

TABLE I
NUMBER OF SAMPLES IN THE YAOWA PORT DATASET

No	Class	Training	Validation	Testing
C1	River	6	6	557
C2	Sea	17	17	1663
C3	Aquaculture	12	12	1211
C4	Building	9	9	946
C5	Pond	13	13	1248
C6	Suaeda salsa	14	14	1344
C7	Farmland	23	23	2263
C8	Mudflat	9	9	908
C9	Road	3	3	300
Total		106	106	10440

TABLE II
NUMBER OF SAMPLES IN THE XINYANG PORT DATASET

No	Class	Training	Validation	Testing
C1	Suaeda	27	28	495
C2	Crops	45	45	808
C3	Sea	151	152	2729
C4	Farmland	90	90	1619
C5	Road	23	23	416
C6	Mudflat	22	22	388
C7	Lake	22	21	387
C8	Salt marsh	19	19	350
C9	Reed	43	43	783
C10	Building	22	22	397
C11	Pond	32	32	581
C12	River	100	100	1793
C13	Aquaculture	219	218	3937
Total		815	815	14683

2) Complexity of BP-BAGCN: To reduce peak memory occupancy, BP-BAGCN divides the image into overlapping spatial blocks. Within each block, feature storage requires $M_F = O(N_k d)$. After pruning, $|\hat{E}_k| = \rho|E_k|$, $0 < \rho < 1$ and adjacency storage under sparse representation requires $M_A = O(|\hat{E}_k|)$. For graph convolution, intermediate propagation features require $M_g = O(N_k d h_l)$. Thus, one block memory becomes $M_{Block} = O(N_k d + \rho|E_k| + N_k d h_l) \approx O(N_k(8\rho + d + dh_l))$, which indicates that memory consumption depends on block size.

For computation, one block requires $T_{BP-BAGCN} \approx O(N_k h_l(8\rho + d))$. Compared with BAGCN, BP-BAGCN reduces peak memory occupancy by restricting graph inference to local subgraphs and removing low-confidence edges, which is more scalable for large hyperspectral scenes.

V. EXPERIMENTS

A. Datasets

To comprehensively evaluate the effectiveness of the proposed method, the Yaowa Port and the Xinyang Port datasets were newly constructed, which provide representative and complex coastal wetland features. To ensure label reliability, the reference maps were generated through a multistage annotation procedure. The expert group conducted field investigations and in situ sampling campaigns within the study regions to collect representative land-cover information. The annotation process further incorporated existing regional ecological surveys and published domain-specific studies to support interpretation of spatial distributions. Experiments on the benchmark Houston dataset are performed to demonstrate the reliability.

1) Yaowa Port: The remote sensing imagery was captured by the AHSI satellite-borne hyperspectral sensor, which was developed by the Shanghai Institute of Technical Physics at the Chinese Academy of Sciences. The sensor is mounted on the GF-5 satellite to observe the ground. Its observation area primarily covers the coastal wetlands of Yancheng in Jiangsu Province, China. The original hyperspectral image contains 330 spectral bands. After removing noisy and water absorption bands, 297 valid bands were retained for experiments. The dataset size is 695 × 556 pixels, and this area contains nine land-cover categories. The number of labeled samples for each category is shown in Table I. The false-color and ground-truth images are shown in Fig. 4. This dataset represents a spectrally complex wetland environment, where subtle differences in vegetation, water bodies, and soil types result in highly similar spectral signatures, making it particularly suitable for evaluating the spectral discrimination capability of classification models.

2) Xinyang Port: Both the Xinyang Port and the Yaowa Port datasets originate from the same satellite sensor. The Xinyang Port dataset encompasses the Yancheng Typical Wetland Ecological Conservation Area. The spatial resolution is 30 m, with an image size of 577 × 496 pixels. After removing noisy and water absorption bands, 297 valid bands are retained for experiments. This area contains 13 basic land-cover categories. The number of labeled samples for each category is shown in Table II. The false-color and ground-truth images are shown in Fig. 5. Compared with the Yaowa Port, this dataset contains a wider variety of land-cover categories and a more complex distribution of land features, providing a more challenging scenario for modeling class discriminability and spatial heterogeneity.

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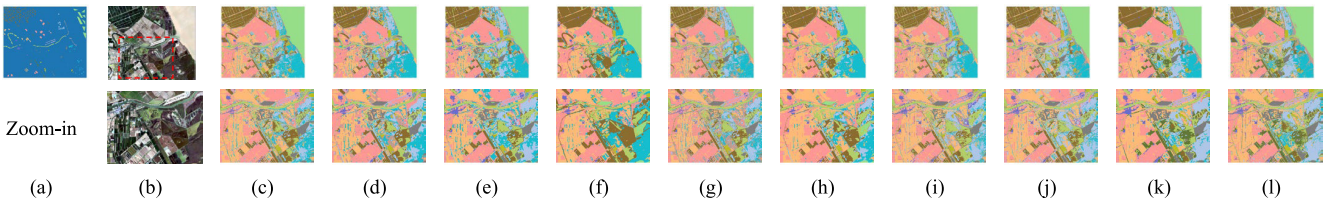


Fig. 5. Classification maps and zoomed-in boundary region on the Xinyang Port dataset. (a) Ground truth. (b) False-color image. (c) miniGCN [57]. (d) GiGCN [53]. (e) GKACN [54]. (f) ViT [20]. (g) SF [23]. (h) SSFTT [58]. (i) MiM [39]. (j) SSMamba [44]. (k) BAGCN. (l) BP-BAGCN.

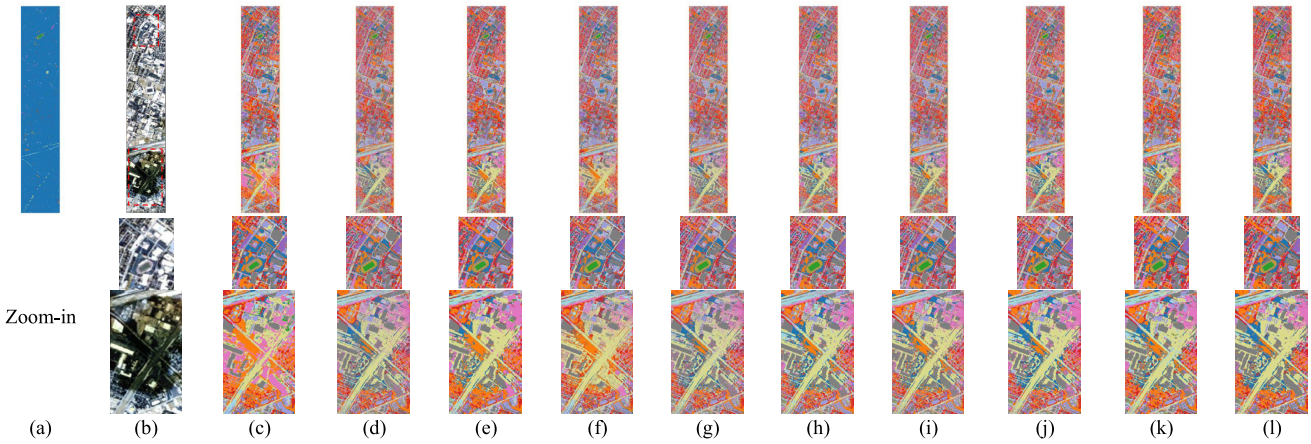


Fig. 6. Classification maps and zoomed-in boundary regions on the Houston 2013 dataset. (a) Ground truth. (b) False-color image. (c) miniGCN [57]. (d) GiGCN [53]. (e) GKACN [54]. (f) ViT [20]. (g) SF [23]. (h) SSFTT [58]. (i) MiM [39]. (j) SSMamba [44]. (k) BAGCN. (l) BP-BAGCN.

TABLE III
NUMBER OF SAMPLES IN THE HOUSTON 2013 DATASET

No	Class	Training	Validation	Testing
C1	Healthy Grass	63	63	1125
C2	Stressed Grass	63	63	1128
C3	Synthetic Grass	35	35	627
C4	Tree	62	62	1120
C5	Soil	62	62	1118
C6	Water	16	16	293
C7	Residential	63	63	1142
C8	Commercial	62	62	1120
C9	Road	63	63	1126
C10	Highway	61	61	1105
C11	Railway	62	62	1111
C12	Parking Lot1	62	62	1109
C13	Parking Lot2	23	23	423
C14	Tennis Court	21	21	386
C15	Running Track	33	33	594
Total		751	751	13527

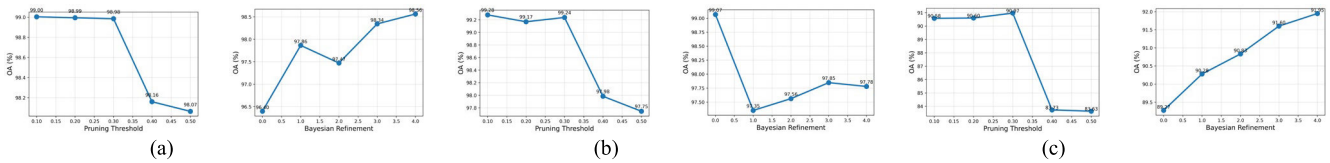
3) Houston 2013: The Houston 2013 hyperspectral dataset is provided by the IEEE Geoscience and Remote Sensing Society (GRSS). The dataset was acquired using an airborne ITRES CASI-1500 sensor over the University of Houston campus and surrounding urban areas in Texas, USA. It comprises 144 bands spanning the 380–1050-nm wavelength range, featuring a spatial resolution of 2.5 m. The image measures 349×1905 pixels and includes 15 land-cover classes. The number of labeled samples for each category is shown in Table III. The false-color and ground-truth images are shown in Fig. 6. This dataset presents a high-resolution urban scene featuring complex man-made structures and fine spatial details. As a widely used public benchmark dataset, its experimental results demonstrate the effectiveness and robustness of the proposed method.

B. Classification

1) Experimental Setup: To evaluate robustness under random sampling, the proposed method is additionally repeated over five independent random seeds. For each dataset, a proportionate number of labeled samples are randomly selected via stratified sampling to form the training set at rates of 1%, 5%, and 5%, respectively. The validation set contains the same number of samples as the training set, while the remaining labeled samples constitute the testing set. The comparison algorithms encompass representative approaches from the latest state-of-the-art models, including graph-based methods such as miniGCN, GiGCN, and GKACN; Transformer-based methods including ViT, SF, and SSFTT; and Mamba-based methods including MiM and SSMamba. All methods are implemented under identical experimental conditions, employing consistent training–testing splits, learning rate scheduling schemes, optimization strategies, and hyperparameter settings to ensure fairness. Evaluation metrics include overall accuracy (OA), average accuracy (AA), category accuracy (CA), and the kappa coefficient, all of which are standard indicators in HSIC.

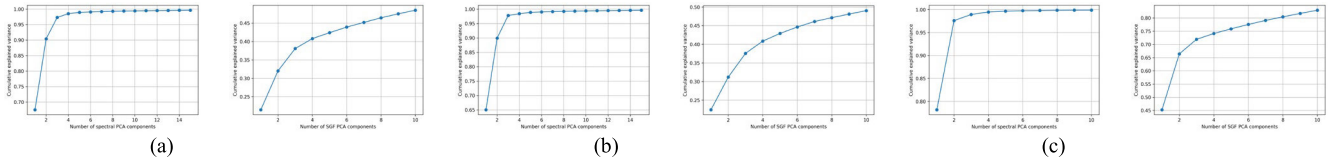
Additionally, experimental analysis is supplemented by multiple visualization outputs. T-SNE feature embedding plots are employed to assess feature separability, while pixel-level spectral curves and SGF response curves illustrate spectral gradient characteristics. Cumulative explained variance plots

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(a) (b) (c)

Fig. 7. Parameter sensitivity analysis of the pruning threshold and Bayesian refinement iterations in BP-BAGCN. (a) Yaowa Port. (b) Xinyang Port. (c) Houston 2013.



(a) (b) (c)

Fig. 8. Spectral and SGF cumulative explained variance as a function of the number of PCA components. (a) Yaowa Port. (b) Xinyang Port. (c) Houston 2013.

for SGF PCA and spectral PCA components evaluate the validity of selected dimensionality levels.

2) Parameters Analysis: Bayesian adaptive edge modeling introduces a probabilistic support and suppression mechanism for graph connectivity. The adaptive weighting strategy effectively mitigates over-smoothing caused by dense graph connections and noise propagation triggered by unreliable neighbors. Cross-dataset experimental stability demonstrates that Bayesian prior parameters and similarity modulation factors can construct robust graph structures without extensive dataset-specific tuning. The Bayesian graph refinement parameters for the three datasets are set as $\alpha = \beta = 1.0$ to impose a weakly informative prior on edge existence, avoiding any artificial bias toward dense or sparse graph structures. Feature similarity is computed using a Gaussian kernel with scale $\sigma = 1.0$ on normalized features. The combined similarity-posterior support is linearly modulated by a power factor $\gamma = 1.0$, enabling the linear propagation of trusted values. To ensure numerical stability during message passing, a minimum edge weight is imposed after Bayesian refinement. Since α and β are adopted as noninformative Beta priors, and σ and γ mainly serve as normalization and similarity scaling coefficients, these parameters are fixed across all datasets throughout the experiments. The sensitivity analysis focuses on the pruning threshold τ and the number of Bayesian refinement iterations, which directly influence graph sparsity, graph refinement behavior, and overall classification performance. The results shown in Fig. 7 indicate that BP-BAGCN remains stable when the pruning threshold is set between 0.1 and 0.3. Classification performance degrades noticeably when the threshold exceeds 0.3 due to over-sparse graph connectivity. The influence of Bayesian refinement iterations has minor performance variations observed across different datasets. Considering both classification accuracy and computational efficiency, the pruning threshold is set to 0.3, and one Bayesian refinement iteration is adopted in all subsequent experiments.

Beyond Bayesian model parameter settings, spectral PCA achieves rapid variance saturation. Observing cumulative explained variance curves reveals how explained variance values change with the number of principal components, indicating significant information redundancy when too many spectral components are used. In the Yaowa Port dataset, 15 spectral principal components nearly capture all information. Similarly, 15 spectral principal components are extracted from the Xinyang Port dataset, while ten spectral principal components suffice for the Houston 2013 dataset. SGF PCA exhibits progressive variance accumulation, preserving discriminative gradient information across multiple components. This indicates that SGF features not only preserve global spectral energy but also maintain intricate spectral variation patterns, which is particularly advantageous for class boundary discrimination, especially among spectrally similar categories. SGF serves as a guiding factor. As shown in Fig. 8, experiments demonstrate that a finite number of SGF bands can effectively enhance the efficiency of image structure construction. Therefore, ten SGF principal components are extracted across all three datasets, providing a good balance between computational cost and classification performance.

In general, the selection of spectral components follows the principle of cumulative variance saturation, typically more than 95%, while the number of SGF components is determined by the tradeoff between structural discrimination and computational efficiency. For other datasets, a similar strategy can be applied by analyzing variance curves. Datasets with higher spectral redundancy typically require 10–20 components, whereas lower-dimensional datasets may require fewer components.

3) Performance: Zoomed-in boundary regions and confusion matrices are provided in Figs. 4–6 and 9 to better analyze local transition areas and class-wise discrimination performance.

According to Fig. 10, the Yaowa Port and Xinyang Port datasets exhibit clear sample distributions in feature space, making classification easier. In contrast, the Houston 2013 dataset shows relatively confused sample distributions in feature space, with certain categories being more difficult to delineate accurately. These visualization results are consistent with the quantitative evaluation presented in Tables IV–VI. Specifically, the more compact intraclass distributions and clearer interclass separation observed for the two coastal wetland datasets correspond to their higher OA, AA, and Kappa values, whereas the relatively overlapping clusters in

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TABLE IV
CLASSIFICATION ACCURACY FOR THE YAOWA PORT DATASET

Class	Graph-based methods			Transformer-based methods			Mamba-based methods		Proposed methods	
	miniGCN	GiGCN	GKACN	ViT	SF	SSFTT	MiM	SSMamba	BAGCN	BP-BAGCN
C1	98.56	**100.00**	98.74	0.36	99.82	98.03	**100.00**	99.64	99.25±0.31	95.66±3.66
C2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00±0.00**	**100.00±0.00**
C3	**100.00**	**100.00**	**100.00**	99.59	99.75	**100.00**	99.83	99.50	99.85±0.18	99.09±1.30
C4	94.19	**95.77**	93.97	85.73	94.71	95.03	93.45	94.71	95.48±3.72	95.56±3.39
C5	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00±0.00**	**100.00±0.00**
C6	99.63	**100.00**	99.85	**100.00**	**100.00**	**100.00**	97.47	99.78	99.91±0.18	99.49±0.70
C7	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00±0.00**	**100.00±0.00**
C8	62.78	83.92	96.15	17.84	96.48	68.39	90.86	95.93	93.81±5.50	**96.92±2.00**
C9	0.00	3.00	55.00	0.00	33.33	0.33	4.67	50.00	**93.13±5.81**	65.93±25.17
OA (%)	93.24	95.43	97.74	83.32	97.26	93.83	95.53	97.62	**98.79±0.70**	97.95±0.55
AA (%)	83.91	86.97	93.75	67.06	91.57	84.64	87.36	93.28	**97.94±0.66**	94.74±2.84
Kappa	0.9214	0.9469	0.9738	0.8046	0.9682	0.9282	0.9481	0.9724	**0.9859±0.0081**	0.9762±0.0064

TABLE V
CLASSIFICATION ACCURACY FOR THE XINYANG PORT DATASET

Class	Graph-based methods			Transformer-based methods			Mamba-based methods		Proposed methods	
	miniGCN	GiGCN	GKACN	ViT	SF	SSFTT	MiM	SSMamba	BAGCN	BP-BAGCN
C1	92.73	**96.36**	95.35	9.29	94.75	80.81	90.91	89.09	96.20±2.91	95.07±2.26
C2	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00±0.00**	**100.00±0.00**
C3	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	**100.00±0.00**	**100.00±0.00**
C4	99.63	**100.00**	99.69	**100.00**	99.63	**100.00**	99.63	98.15	99.91±0.11	98.88±0.41
C5	80.53	77.64	94.47	0.00	79.09	0.00	70.43	85.10	**98.27±1.38**	88.65±3.07
C6	46.13	65.21	85.57	0.00	76.80	25.77	62.37	81.19	**97.94±1.38**	87.68±3.34
C7	84.24	82.69	91.73	0.00	71.58	11.11	70.54	92.25	98.71±2.58	**99.38±1.24**
C8	90.29	93.43	98.57	77.71	93.43	90.57	97.14	95.71	**99.66±0.21**	99.03±0.78
C9	96.81	95.15	99.87	**100.00**	96.04	**100.00**	98.60	98.08	99.51±0.84	98.70±0.73
C10	62.72	80.86	98.49	82.62	56.42	72.54	88.16	87.66	**98.64±0.72**	86.25±3.24
C11	0.00	0.00	8.09	0.00	0.00	0.00	0.00	1.89	**94.94±2.14**	84.41±5.15
C12	78.69	80.03	97.66	52.20	79.98	74.68	94.42	95.93	**99.99±0.02**	99.34±0.47
C13	99.57	99.85	98.32	**100.00**	99.80	**100.00**	98.76	98.83	99.61±0.32	98.12±0.49
OA (%)	89.24	90.50	94.60	78.04	89.82	84.20	91.61	93.07	**99.35±0.13**	97.38±0.27
AA (%)	79.33	82.40	89.83	55.53	80.58	65.81	82.38	86.45	**98.72±0.48**	95.04±0.86
Kappa	0.8718	0.8870	0.9362	0.7337	0.8788	0.8104	0.9005	0.9180	**0.9924±0.0015**	0.9693±0.0032

TABLE VI
CLASSIFICATION ACCURACY FOR THE HOUSTON 2013 DATASET

Class	Graph-based methods			Transformer-based methods			Mamba-based methods		Proposed methods	
	miniGCN	GiGCN	GKACN	ViT	SF	SSFTT	MiM	SSMamba	BAGCN	BP-BAGCN
C1	85.42	93.33	91.20	85.42	93.51	96.00	93.16	96.98	**98.67±1.17**	98.35±1.16
C2	89.72	93.88	95.48	93.17	94.95	**98.58**	97.25	98.32	97.41±1.61	98.01±1.27
C3	94.26	99.36	96.97	99.04	99.68	99.68	99.20	**99.84**	98.28±0.89	98.41±0.77
C4	93.21	94.02	93.93	93.12	94.20	94.55	92.95	95.45	94.96±2.52	**95.93±2.38**
C5	99.55	96.51	**99.73**	97.85	97.05	98.39	97.23	98.57	99.59±0.34	99.70±0.26
C6	86.01	89.76	90.44	87.03	88.74	94.54	86.69	88.40	95.56±2.85	**95.70±2.26**
C7	88.70	74.08	86.25	78.81	75.22	84.76	83.10	90.19	88.90±3.37	**90.88±2.71**
C8	53.21	82.77	70.71	57.68	83.75	83.39	**85.27**	82.50	83.00±2.76	82.55±1.76
C9	67.94	76.11	76.55	71.49	73.00	82.50	77.89	79.84	86.06±2.85	**86.48±0.94**
C10	60.18	73.48	78.10	66.33	73.57	84.98	84.52	89.95	**91.64±3.48**	88.51±3.96
C11	72.01	65.35	79.30	69.94	68.50	85.87	72.10	82.90	**88.06±3.08**	**88.06±3.48**
C12	75.74	69.43	76.56	70.06	71.87	84.04	79.89	86.93	**90.30±2.03**	88.76±2.99
C13	30.02	21.99	69.27	15.13	25.06	28.84	24.82	27.42	76.45±4.27	**77.83±2.65**
C14	92.75	82.12	98.70	73.32	86.79	89.90	86.53	96.63	**98.76±0.81**	98.50±1.09
C15	98.82	99.16	97.64	**99.49**	98.99	98.99	98.99	98.65	99.12±0.94	99.12±0.48
OA (%)	79.28	81.79	85.97	78.37	82.55	88.46	85.59	89.18	**92.27±0.65**	92.21±0.51
AA (%)	79.17	80.76	86.72	77.19	81.66	87.00	83.97	87.50	**92.45±0.52**	**92.45±0.38**
Kappa	0.7757	0.8029	0.8482	0.7658	0.8112	0.8751	0.8440	0.8829	**0.9164±0.0070**	0.9158±0.0055

the Houston 2013 dataset are consistent with its comparatively lower overall classification accuracy and reflect the greater difficulty of discriminating spectrally similar urban categories.

Figs. 11–13 show the spectral curves of randomly selected pixels from each category across the three datasets, alongside their respective SGF plots. The blue curves represent the original spectral data and reveal strong similarities between certain categories. The orange SGF response lines highlight local spectral variations and gradient changes, extracting differential information not observable in the original spectral bands that can be used for discrimination. SGF effectively improves the separability of features in diverse samples. Samples with similar spectral signatures but different local spectral transition characteristics become more distinguishable,

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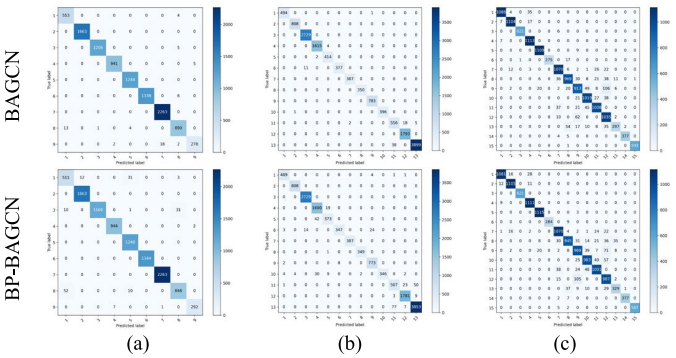


Fig. 9. Confusion matrices of BAGCN and BP-BAGCN. (a) Yaowa Port. (b) Xinyang Port. (c) Houston 2013.

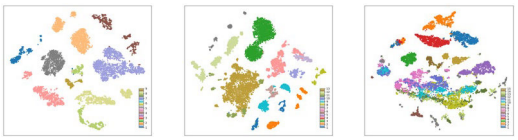


Fig. 10. Illustration of T-SNE feature distributions. (a) Yaowa Port. (b) Xinyang Port. (c) Houston 2013.

leading to improved feature compactness and reduced confusion among spectrally similar categories.

Based on the classification performance metrics, the proposed BAGCN and BP-BAGCN consistently achieve the highest OA, AA, and Kappa values across all three datasets. For the Yaowa Port dataset, as shown in Table IV, BAGCN attains the highest OA of 98.79%, followed by BP-BAGCN with an OA of 97.95%. Compared with other state-of-the-art methods, both approaches demonstrate outstanding performance. Notably, BAGCN achieves a 15.47% improvement in OA over ViT, a 5.55% improvement in OA over miniGCN, and a 3.26% improvement in OA over MiM. Furthermore, the proposed methods yield average Kappa values of 0.9859 and 0.9762, respectively, indicating exceptionally high consistency in high-precision classification results across all classes.

For the second dataset, as shown in Table V, BAGCN again achieves the highest OA of 99.35%, significantly outperforming Transformer-based methods. Specifically, the proposed method achieves an OA approximately 10%–20% higher than Transformer-based approaches and about 5%–10% higher than graph-based and Mamba methods. Its Kappa coefficient significantly outperforms other methods. Analysis of classification across categories reveals that other methods are prone to misclassifying the eleventh category, adversely affecting overall classification performance. In contrast, the proposed methods demonstrate robust classification across all categories with stable results.

For the two coastal wetland datasets, the most evident gains appear in categories with strong spectral overlap and heterogeneous spatial distribution. Categories such as river, pond, aquaculture, mudflat, salt marsh, reed, and suaeda exhibit highly similar spectral responses and complex transition boundaries. Traditional graph models and global attention mechanisms tend to introduce neighborhood ambiguity in these regions. By incorporating SGF features, the proposed framework captures subtle spectral transition patterns rather than relying solely on absolute reflectance values, improving discrimination among spectrally adjacent wetland classes. Bayesian graph refinement selectively strengthens reliable local relationships and suppresses uncertain message propagation, which is particularly beneficial in fragmented coastal environments containing mixed pixels and irregular boundaries. These improvements are reflected in the classification maps, where smoother homogeneous regions and clearer object boundaries are observed compared with competing methods. The visual reduction of fragmented boundary errors is consistent with the higher class-wise accuracies achieved for river, pond, aquaculture, mudflat, reed, and salt marsh, indicating that the proposed SGF-guided Bayesian graph refinement effectively reduces confusion among neighboring wetland categories.

For the third dataset, as shown in Table VI, classification performance is relatively consistent across methods. BAGCN achieves slightly better results than BP-BAGCN at 92.27%, outperforming the best graph-based competitor GKACN by 6.30%, the best Transformer-based competitor SSFTT by 3.81%, and the best Mamba-based competitor SSMamba by 3.09%. The proposed methods also achieve excellent Kappa values of 0.9164 and 0.9158, demonstrating strong robustness under complex class distributions.

For the Houston 2013 dataset, improvements are mainly observed in categories with complex urban structures and strong local context dependency. Classes including residential, commercial, road, highway, railway, and parking lot present substantial intraclass variability and interclass similarity due to mixed materials and varying spatial layouts. Compared with Transformer-based and Mamba-based methods, the proposed approach achieves more stable performance because graph refinement explicitly models local relational consistency while SGF preserves fine spectral variation information. The results suggest that adaptive graph updating complements global representation learning by reducing unreliable information aggregation near object boundaries. Consistent with this observation, the classification maps contain fewer isolated misclassified pixels around roads, residential areas, commercial regions, and parking lots than competing methods. This visual improvement agrees with the higher class-wise accuracies reported in Table VI, indicating reduced confusion among adjacent urban classes with similar spectral characteristics.

4) Ablation Analysis: To investigate the influence of the initial graph topology, we additionally compare the proposed framework using the conventional 8-neighborhood graph and a spectral kNN graph with k = 8. The comparison is intended to evaluate the robustness of the proposed Bayesian graph refinement with respect to different graph initializations rather than to optimize graph construction itself. As shown in Table VII, both the 8-neighborhood graph and the spectral kNN graph produce competitive classification performance across all three datasets. Although the eight-neighborhood graph consistently achieves slightly higher accuracy under the same graph sparsity, the overall performance trends remain unchanged. These results suggest that the proposed

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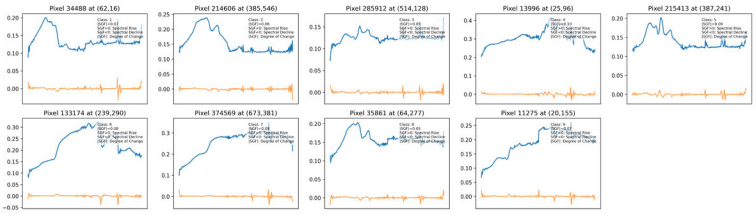


Fig. 11. Spectral curves of randomly selected pixels from each category in the Yaowa Port dataset.

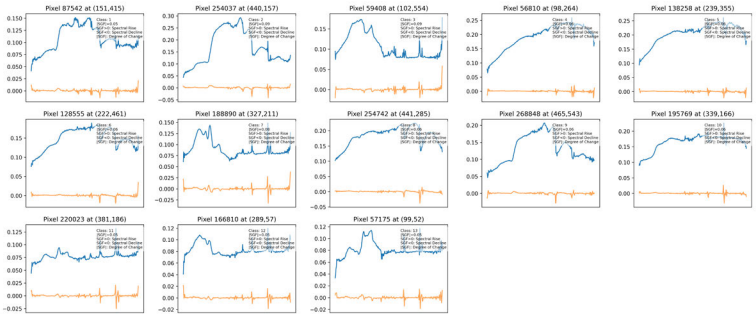


Fig. 12. Spectral curves of randomly selected pixels from each category in the Xinyang Port dataset.

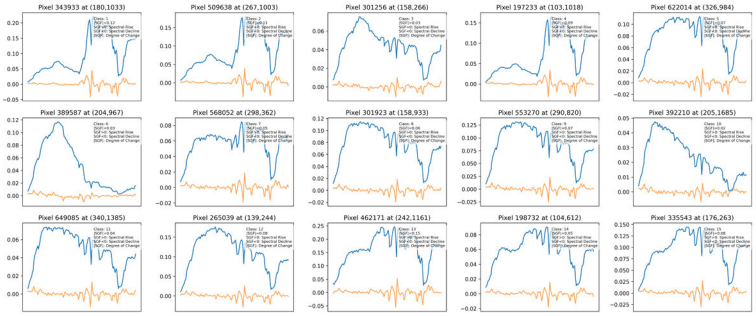


Fig. 13. Spectral curves of randomly selected pixels from each category in the Houston 2013 dataset.

TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT INITIAL GRAPH CONSTRUCTION STRATEGIES

Datasets	Initial Graph		OA(%)	AA(%)	Kappa	Time(s)
Yaowa Port	8-neighborhood	BAGCN	98.79±0.70	97.94±0.66	0.9859±0.0081	177.71±6.59
		BP-BAGCN	97.95±0.55	94.74±2.84	0.9762±0.0064	167.58±7.11
	Spectral kNN (k=8)	BAGCN	97.61±0.58	96.55±1.17	0.9723±0.0067	310.49±10.04
		BP-BAGCN	95.69±0.56	88.33±0.47	0.9499±0.0065	292.33±7.68
Xinyang Port	8-neighborhood	BAGCN	99.35±0.13	98.72±0.48	0.9924±0.0015	125.50±6.81
		BP-BAGCN	97.38±0.27	95.04±0.86	0.9693±0.0032	120.29±6.15
	Spectral kNN (k=8)	BAGCN	97.56±0.18	95.43±0.43	0.9715±0.0021	217.86±8.29
		BP-BAGCN	97.28±0.26	96.25±0.65	0.9682±0.0030	197.62±6.64
Houston 2013	8-neighborhood	BAGCN	92.27±0.65	92.45±0.52	0.9164±0.0070	291.27±6.65
		BP-BAGCN	92.21±0.51	92.45±0.38	0.9158±0.0055	290.21±3.76
	Spectral kNN (k=8)	BAGCN	87.42±0.40	86.68±0.34	0.8639±0.0043	593.53±1.69
		BP-BAGCN	86.89±0.90	86.25±0.82	0.8581±0.0097	572.46±5.65

SGF-guided Bayesian adaptive graph refinement is not critically dependent on a specific graph initialization, while the initial topology mainly serves as the starting point for subsequent graph optimization.

To validate the contribution of each module to the enhancement of model performance, this article designs systematic ablation experiments. Full model: the complete framework BP-BAGCN. Model A: the full model without the SGF features. Model B: the proposed BAGCN. Model C: enhances node representations with SGF features. Model D: represents the baseline GCN without SGF features or Bayesian graph refinement. Table VIII presents the model architectures and the OA comparisons across them on the Yaowa Port, Xinyang Port, and Houston 2013 datasets.

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TABLE VIII
ABLATION ANALYSIS

Model	SGF	Bayesian Adaptive	Block-wise	Datasets		
				Yaowa Port	Xinyang Port	Houston 2013
Full	√	√	√	98.19	97.27	**92.67**
A	×	√	√	97.73	95.84	84.70
B	√	√	×	**99.10**	**99.12**	92.26
C	√	×	×	93.76	95.22	88.69
D	×	×	×	90.31	91.81	83.14

A comparison of model A with the full model indicates that, upon the removal of SGF and the retention of raw spectral features alone, the OA rates for the three datasets undergo a decline, from 98.19% to 97.73%, 97.27% to 95.84%, and 92.67% to 84.70%, respectively. The Houston dataset exhibits the most substantial decline, with a decrease of 7.97%. It is evident that reliance on raw spectral information alone is inadequate for achieving a comprehensive representation of the characteristics. Model B, which removes the block pruning mechanism, achieves OA of 99.10%, 99.12%, and 92.26% across three datasets. This demonstrates that the block pruning mechanism effectively reduces the interference of redundant connections in the graph inference process while ensuring performance stability. It enhances the sparsity and shows that reliance on raw spectral information alone is inadequate for achieving a comprehensive representation of the characteristics. Model B, which removes the block pruning mechanism, achieves OA of 99.10%, 99.12%, and 92.26% across three datasets. This demonstrates that the block pruning mechanism effectively reduces the interference of redundant connections in the graph inference process while ensuring performance stability. It enhances the sparsity and effectiveness of graph structure representation, thereby improving overall modeling quality and generalization capability. Model C retains only the SGF and basic graph structure, showing respective decreases of 4.43%, 2.05%, and 3.98% compared to the full model. This result indicates that a fixed graph structure struggles to address noise interference, spectral aliasing, and local structural instability in hyperspectral images. This further demonstrates that Bayesian adaptive graph modeling, by incorporating probabilistic edge weights and uncertainty modeling mechanisms, enables dynamic adjustment of graph connectivity relationships and reliable evaluation. Model D, which removes all modules simultaneously, exhibits significant performance degradation. This indicates that traditional graph convolutional frameworks relying solely on raw spectral features and fixed graph structures struggle to effectively capture nonlinear categorical structures and spatial spectral correlations in complex hyperspectral scenes.

VI. CONCLUSION

This article presented an SGF-guided BAGCN framework for HSIC. By introducing the SGF as a physically interpretable prior, the proposed method provides principled guidance for graph construction and message propagation beyond conventional spectral similarity. A Bayesian adaptive graph refinement strategy is further developed to dynamically optimize edge connectivity based on both spectral gradient consistency and model-driven class evidence, enabling uncertainty-aware and data-adaptive graph structure learning. To improve scalability, a block-wise inference scheme combined with posterior-based edge pruning is designed to reduce memory consumption and computational complexity while preserving essential structural information. Extensive experiments were conducted on three datasets, including two newly constructed GF-5 hyperspectral datasets covering complex coastal wetland environments. The diversity of these datasets, in terms of spectral variability, spatial heterogeneity, and class distribution, validates the effectiveness and generalization capability of the proposed approach in practical scenarios. Specifically, BAGCN improves the OA by 15.47% over ViT on the Yaowa Port dataset, by approximately 5%–10% higher than graph-based and Mamba methods on the Xinyang Port dataset, and by 6.30% over the best graph-based method on the Houston 2013 dataset.

This work establishes a unified probabilistic graph learning paradigm that integrates physical spectral priors with Bayesian inference for adaptive graph convolution, providing an effective and scalable solution for HSIC. Owing to its modular design and data-adaptive nature, the proposed framework also shows strong potential for extension to other remote sensing tasks, such as multimodal data fusion and change detection, as well as to graph-based learning problems in other domains. Nevertheless, several limitations remain. Since posterior graph refinement partially depends on model-derived soft class agreement, inaccurate early predictions may influence graph evolution. In the current framework, Bayesian confidence updating and posterior-based edge pruning partially alleviate error accumulation by suppressing low-confidence connections. In addition, although the proposed block-wise strategy substantially improves scalability for large hyperspectral scenes, local partitioning may weaken long-range dependency modeling and introduce boundary inconsistency across blocks. These limitations do not invalidate the effectiveness demonstrated in the experiments but indicate opportunities for further improving robustness, uncertainty modeling, scalability, and generalization. In future work, the current graph-level Bayesian refinement may be extended toward uncertainty-aware prediction and calibrated uncertainty quantification for more reliable decision-making. Meanwhile, overlapping block partitioning, boundary-aware graph stitching, cross-block message passing, and hierarchical global–local graph construction may be explored to better preserve global topology while maintaining computational efficiency. In addition, the integration of self-supervised learning with adaptive graph learning may be a promising direction to further enhance feature generalization under limited labeled data.

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newly constructed GF-5 coastal wetland datasets are welcome to contact the corresponding author regarding data access.

REFERENCES

[1] T. Zhao et al., "Artificial intelligence for geoscience: Progress, challenges, and perspectives," *Innov.*, vol. 5, no. 5, Sep. 2024, Art. no. 100691, doi: 10.1016/j.xinn.2024.100691.

[2] L. Yinnian, "Development of hyperspectral imaging remote sensing technology," *Nat. Remote Sens. Bull.*, vol. 25, no. 1, pp. 439–459, 2021, doi: 10.11834/jrs.20210283.

[3] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004, doi: 10.1109/TGRS.2004.831865.

[4] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, Jan. 2023, Art. no. 0025, doi: 10.34133/remotesensing.0025.

[5] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag., Replaces Newslett.*, vol. 1, no. 2, pp. 6–36, Jun. 2013, doi: 10.1109/MGRS.2013.2244672.

[6] X. Li, Z. Li, H. Qiu, G. Hou, and P. Fan, "An overview of hyperspectral image feature extraction, classification methods and the methods based on small samples," *Appl. Spectrosc. Rev.*, vol. 58, no. 6, pp. 367–400, Jul. 2023, doi: 10.1080/05704928.2021.1999252.

[7] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005, doi: 10.1109/TGRS.2004.842481.

[8] J. Xia, P. Ghamisi, N. Yokoya, and A. Iwasaki, "Random forest ensembles and extended multiextinction profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 202–216, Jan. 2018, doi: 10.1109/TGRS.2017.2744662.

[9] G. Mountrakis, J. Im, and C. Ogole, "Support vector machines in remote sensing: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 3, pp. 247–259, May 2011, doi: 10.1016/j.isprsjprs.2010.11.001.

[10] J. Peng et al., "Low-rank and sparse representation for hyperspectral image processing: A review," *IEEE Geosci. Remote Sens. Mag., Replaces Newslett.*, vol. 10, no. 1, pp. 10–43, Mar. 2022, doi: 10.1109/MGRS.2021.3075491.

[11] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011, doi: 10.1109/TGRS.2011.2129595.

[12] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification via kernel sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 1, pp. 217–231, Jan. 2013, doi: 10.1109/TGRS.2012.2201730.

[13] H. Su, Y. Yu, Q. Du, and P. Du, "Ensemble learning for hyperspectral image classification using tangent collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 3778–3790, Jun. 2020, doi: 10.1109/TGRS.2019.2957135.

[14] H. Su, F. Shao, Y. Gao, H. Zhang, W. Sun, and Q. Du, "Probabilistic collaborative representation based ensemble learning for classification of wetland hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509517, doi: 10.1109/TGRS.2023.3267638.

[15] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005, doi: 10.1109/TGRS.2004.842478.

[16] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008, doi: 10.1109/TGRS.2008.922034.

[17] D. Huang, C. Shan, M. Ardabilian, Y. Wang, and L. Chen, "Local binary patterns and its application to facial image analysis: A survey," *IEEE Trans. Syst., Man, Cybern., C, Appl. Rev.*, vol. 41, no. 6, pp. 765–781, Nov. 2011, doi: 10.1109/TSMCC.2011.2118750.

[18] L. Fang, S. Li, X. Kang, and J. A. Benediktsson, "Spectral–spatial classification of hyperspectral images with a superpixel-based discriminative sparse model," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4186–4201, Aug. 2015, doi: 10.1109/TGRS.2015.2392755.

[19] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017, doi: 10.1109/TGRS.2017.2693346.

[20] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[21] C. Zhou, Z. He, L. Wang, J. Ren, and A. Plaza, "PLGMamba: A new progressive local–global state-space model for hyperspectral image super-resolution," *J. Remote Sens.*, vol. 6, Jan. 2026, Art. no. 1027, doi: 10.34133/remotesensing.1027.

[22] H. Zheng, H. Su, Z. Wu, M. E. Paoletti, and Q. Du, "Graph convolutional network with relaxed collaborative representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5533513, doi: 10.1109/TGRS.2024.3468269.

[23] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615, doi: 10.1109/TGRS.2021.3130716.

[24] Y. Peng, Y. Zhang, B. Tu, Q. Li, and W. Li, "Spatial–spectral transformer with cross-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537415, doi: 10.1109/TGRS.2022.3203476.

[25] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715, doi: 10.1109/TGRS.2022.3171551.

[26] L. Song, Z. Feng, S. Yang, X. Zhang, and L. Jiao, "Interactive spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8589–8601, Sep. 2024, doi: 10.1109/TCSVT.2024.3386578.

[27] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513, doi: 10.1109/TGRS.2022.3186400.

[28] Z. Chen, S. Kumar Roy, H. Gao, Y. Ding, and B. Zhang, "SiT: Scale-interaction transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522617, doi: 10.1109/TGRS.2025.3598290.

[29] G. Chen, L. Yang, T. Wu, L. Liang, X. Li, and J. Li, "MSSCFormer: Multigranularity spatial–spectral convolution transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5518918, doi: 10.1109/TGRS.2025.3589190.

[30] J. Bai et al., "Hyperspectral image classification based on multibranch attention transformer networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535317, doi: 10.1109/TGRS.2022.3196661.

[31] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817, doi: 10.1109/TGRS.2024.3377610.

[32] X. Zhang, Y. Su, L. Gao, L. Bruzzone, X. Gu, and Q. Tian, "A lightweight transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517617, doi: 10.1109/TGRS.2023.3297858.

[33] S. Mei, C. Song, M. Ma, and F. Xu, "Hyperspectral image classification using group-aware hierarchical transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5539014, doi: 10.1109/TGRS.2022.3207933.

[34] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214, doi: 10.1109/TGRS.2024.3419266.

[35] S. K. Roy, A. Deria, C. Shah, J. M. Haut, Q. Du, and A. Plaza, "Spectral–spatial morphological attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503615, doi: 10.1109/TGRS.2023.3242346.

[36] W.-T. Zhang, Y. Bai, S.-D. Zheng, J. Cui, and Z.-Z. Huang, "Tensor transformer for hyperspectral image classification," *Pattern Recognit.*, vol. 163, Jul. 2025, Art. no. 111470, doi: 10.1016/j.patcog.2025.111470.

[37] S. Jia, Y. Wang, S. Jiang, and R. He, "A center-masked transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5510416, doi: 10.1109/TGRS.2024.3369075.

[38] C. Wang, J. Huang, M. Lv, H. Du, Y. Wu, and R. Qin, "A local enhanced mamba network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104092, doi: 10.1016/j.jag.2024.104092.

[39] W. Zhou, S.-I. Kamata, H. Wang, M. S. Wong, and H. Hou, "Mamba-in-Mamba: Centralized Mamba-cross-scan in tokenized mamba model for hyperspectral image classification," *Neurocomputing*, vol. 613, Jan. 2025, Art. no. 128751, doi: 10.1016/j.neucom.2024.128751.

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[40] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "3DSS-Mamba: 3D-spectral–spatial Mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5534216, doi: 10.1109/TGRS.2024.3472091.

[41] Y. He, B. Tu, P. Jiang, B. Liu, J. Li, and A. Plaza, "IGroupSS-Mamba: Interval group spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538817, doi: 10.1109/TGRS.2024.3502055.

[42] M. Ahmad et al., "Spatial–spectral morphological mamba for hyperspectral image classification," *Neurocomputing*, vol. 636, Jul. 2025, Art. no. 129995, doi: 10.1016/j.neucom.2025.129995.

[43] H. Zhang, X. Xu, S. Li, and A. Plaza, "Wavelet decomposition-based spectral–spatial mamba network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5518817, doi: 10.1109/TGRS.2025.3590154.

[44] J. Sheng, J. Zhou, J. Wang, P. Ye, and J. Fan, "DualMamba: A lightweight spectral–spatial mamba-convolution network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501415, doi: 10.1109/TGRS.2024.3516817.

[45] Y. He, B. Tu, B. Liu, J. Li, and A. Plaza, "HSI-MFormer: Integrating mamba and transformer experts for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5621916, doi: 10.1109/TGRS.2025.3564167.

[46] L. Liang et al., "DBMLLA: Double-branch Mamba-like linear attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5524315, doi: 10.1109/TGRS.2025.3605798.

[47] A. Yang, M. Li, Y. Ding, L. Fang, Y. Cai, and Y. He, "GraphMamba: An efficient graph structure learning vision mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5537414, doi: 10.1109/TGRS.2024.3493101.

[48] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020, doi: 10.1109/TGRS.2020.2973363.

[49] S. Wan, C. Gong, P. Zhong, S. Pan, G. Li, and J. Yang, "Hyperspectral image classification with context-aware dynamic graph convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 597–612, Jan. 2021, doi: 10.1109/TGRS.2020.2994205.

[50] B. Yang, F. Cao, and H. Ye, "A novel method for hyperspectral image classification: Deep network with adaptive graph structure integration," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523512, doi: 10.1109/TGRS.2022.3150349.

[51] J. Chen, L. Jiao, X. Liu, L. Li, F. Liu, and S. Yang, "Automatic graph learning convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520716, doi: 10.1109/TGRS.2021.3135084.

[52] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "Multilevel superpixel structured graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516115, doi: 10.1109/TGRS.2021.3112586.

[53] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1157–1171, Jan. 2024, doi: 10.1109/TNNLS.2022.3182715.

[54] H. Zheng, H. Su, Z. Wu, Y. Chen, and Q. Du, "Graph Kolmogorov–Arnold convolutional network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 14738–14751, 2025, doi: 10.1109/JSTARS.2025.3575805.

[55] Y. Wang, Z. Xue, M. Jia, Z. Liu, and H. Su, "Hypergraph convolutional network with multiple hyperedges fusion for hyperspectral image classification under limited samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526318, doi: 10.1109/TGRS.2024.3440271.

[56] J.-Y. Yang, J.-H. Yang, H.-C. Li, S. Mei, Q. Du, and A. Plaza, "Tensorized high-order hypergraph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5527316, doi: 10.1109/TGRS.2025.3613347.

[57] J. Shekhar, M. Ramya, A. Chandrasekar, S. Nithyanandh, P. Deepa, and A. Madhan, "MiniGCN: Mini-batch graph convolutional networks for hyperspectral image classification," in *Proc. Global Conf. Inf. Technol. Commun. Netw. (GITCON)*, Aug. 2025, pp. 1–6, doi: 10.1109/gitcon65266.2025.11378144.

[58] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214, doi: 10.1109/TGRS.2022.3144158.

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